

The flash tool is designed to facilitate the development, programming, and debugging of applications on Nordic Semiconductor’s nRF series chips. These chips are widely used in IoT, wearable technology, and other wireless communication applications.

Key Features

- Broad compatibility with multiple nRF series SoCs for flexible development
- Seamless firmware programming, bootloader updates, and device configuration
- Reliable debugging with real-time code inspection and error tracking
- Cross-platform usability for developers working on different systems
- Smooth integration with Nordic SDKs and supporting development tools
- Enhanced workflow efficiency through breakpoint control and stepwise debugging
- User-friendly operation suitable for both beginners and advanced developers



Fig: Flash Tool

Technical Specifications

Parameter	Value
Supported Devices	Compatibility with various nRF series SoCs (e.g., nRF51, nRF52, nRF53 series)
Programming Interface	Interfaces such as JTAG, SWD (Serial Wire Debug), or USB
Software Compatibility	Support for various Integrated Development Environments (IDEs) like SEGGER Embedded Studio, Keil, and others
Operating System Support	Availability for multiple operating systems such as Windows, macOS, and Linux
Programming Modes	Ability to program and flash firmware, update bootloaders, and configure device settings
Debugging Capabilities	Support for breakpoints, single-stepping, and inspection of memory and registers
Integration	Possible integration with other development tools like nRF Connect SDK, nRF Command Line Tools, and nRF Connect for Desktop

Applications

- Firmware Development – writing and deploying firmware for nRF-based devices, essential for creating custom applications in Bluetooth LE, ANT, Thread, Zigbee, and other wireless protocols
- Debugging – providing tools to debug and troubleshoot firmware issues, enabling developers to identify and fix bugs efficiently
- Testing – facilitating testing of wireless applications to ensure they meet required specifications and performance metrics

- Production Programming – used in manufacturing environments to program multiple devices with the final firmware version
- Bootloader Updates – updating bootloaders on nRF devices to support new features or fix issues
- Configuration – setting device parameters, calibrating sensors, and adjusting wireless settings